Unproved candidate in the ordinary singular-homology model. This statement translates to a Lean proposition; it is not a proof of the paper's theorem. The reverse inclusion and the closed-cover homology audit remain open.

**Result 1 (interlevel two image candidate).** [Bendich, Edelsbrunner, Morozov, and Patel, Theorem 4]
Let $K$ be a type. Let *field* be a field structure on $K$. Let $p$ be a natural number. Let $H = H_p(-; K)$. Let $P$ be interlevel problem data. Let $r$ be a real number. Assume $P$.space is compact (`compactness of P.space`). Assume $P$.interval.lower $<$ $P$.interval.upper (`endpoints`). Assume $r \geq 0$ (`radius`). Assume $X_r(P)$ is a nested pair (`nesting`). Let $E$ be certificates for the two extreme perturbations. Then

$$U_r(P; H) = I_r^+(P; H, E) \cap I_r^-(P; H, E).$$

Adapted from the cited source. Scope: unproved singular-homology candidate.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.